



PROJECT REPORT

RESTURANT DISH MANAGEMENT SYSTEM

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INTRODUCTION :

The current application is a console-based Restaurant Management System implemented by applying Object-Oriented Programming in the language of C++. In this software system, we can observe how a basic program can manage the details of menu items in a restaurant efficiently while applying the basic aspects of OOP like inheritance, encapsulation, polymorphism, and abstraction.

FEATURES :

We start from the basic concept of a dish in the base class known as Dish. This base class is responsible for creating important functions to get and set the details of a dish like the name and price. Furthermore, the class contains a function which prints all the details of the dish. In continuation with this, various derived classes of the base class "Dish" are created which classify dishes according to their category and type.

In order to further categorize the objects, special classes, which consist of type and courses, are used in the application by implementing multiple inheritance. In this way, the application can categorize the food items into more complex groups without having to create new classes. Furthermore, virtual inheritance was introduced to solve the diamond problem, ensuring that there is only one instance of the base class at any point in time.

Moreover, the application has created a class named Cuisine. This is a class, which can hold an array of dishes belonging to a certain cuisine type. It uses a dynamic data structure to hold the menu list and performs functions of adding new items to the list and printing the list. The use of global structures and pointers is done to enable communication between different parts of the system, which will enable menu items to be dynamically selected and shown.

FUTURE SCOPE :

For now, the program runs on the console platform; however, due to the nature of its design, it is highly modularized, making it possible to add new features in the future, such as a graphical user interface or even access through the web interface. On the whole, the project is an illustration of how the concepts of OOP can be used effectively in practice.
